SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	AIDS/2024	**Date:**	17/08/2026

Code of Conduct

I M.KASIF JALAL (192424214)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: KASIF JALAL M

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A language processing system must represent sentence structure for grammar checking and syntactic analysis. The developers are deciding between Context-Free Grammar (CFG) trees and dependency parsing structures. Analyze how these two approaches differ in representing sentence structure and justify which is more effective for capturing relationships between words.

Program:

```
sentence="The student reads the book."
words=sentence.split()
print("Sentence:",sentence)
print("Dependency Representation:")
print("reads -> student : subject")
print("reads -> book : object")
print("student -> The : determiner")
print("book -> the : determiner")
```

Output:

```
Sentence: The student reads the book.
Dependency Representation:
reads -> student : subject
reads -> book : object
student -> The : determiner
book -> the : determiner
```

2. A real-time application needs to parse user input efficiently, even when sentences are incomplete or ambiguous. The system currently uses top-down parsing, but an alternative is Earley parsing.
Analyze the differences between these parsing techniques and reason which is more suitable for such dynamic input conditions.

Program:

```
sentence="Book a flight to Delhi"
words=sentence.split()
print("Input:",sentence)
print("Parsing Strategy: Earley")
for i,word in enumerate(words,1):
    print("State",i,": processed ->",word)
print("Result: Complete sentence structure identified")
```

Output:

```
Input: Book a flight to Delhi
Parsing Strategy: Earley
State 1 : processed -> Book
State 2 : processed -> a
State 3 : processed -> flight
State 4 : processed -> to
State 5 : processed -> Delhi
Result: Complete sentence structure identified
```

3. An NLP model must handle ambiguity in sentences such as “She saw the man with a telescope.” The

designers are considering CFG, PCFG, and neural parsing techniques.
Analyze how these approaches differ in handling ambiguity and reason which method is most effective for real-world applications.

Program:
```
parses={"She uses telescope":0.70,"Man has telescope":0.30}
best=max(parses,key=parses.get)
print("Sentence: She saw the man with a telescope.")
print("Possible interpretations:")
for k,v in parses.items():
    print(k,"->",v)
print("Selected interpretation:",best)
```

Output:

```
Sentence: She saw the man with a telescope.
Possible interpretations:
She uses telescope -> 0.7
Man has telescope -> 0.3
Selected interpretation: She uses telescope
```

4. A grammar system needs to correctly handle subject–verb agreement and verb argument structures. The developers are comparing feature structures and subcategorization frames.
Analyze how these approaches differ and justify which provides better support for enforcing grammatical correctness.

Program:
```
subject={"word":"student","number":"singular"}
verb={"word":"reads","number":"singular"}
if subject["number"]==verb["number"]:
    print("Agreement: Correct")
else:
    print("Agreement: Error")
frames={"eat":"NP","give":"NP NP","depend":"PP"}
print("prescribe -> NP")
print("give -> NP NP")
```

Output:
```
Agreement: Correct
prescribe -> NP
give -> NP NP
```

5. A large-scale NLP system must perform fast and accurate dependency parsing on big datasets. The choice is between transition-based parsing and graph-based parsing.
Analyze the differences between these methods in terms of decision-making and performance, and justify which is more suitable for large-scale applications.

Program:
```
sentence="The student reads the book."
words=sentence.split()
print("Sentence:",sentence)
print("Transition-Based Parsing:")
```

```
for word in words:
    print("SHIFT ->",word)
print("Dependency tree created using local transitions.")
print("Graph-Based Parsing:")
print("Possible dependency edges evaluated.")
print("Best global dependency tree selected.")
```

Output:

```
Sentence: The student reads the book.
Transition-Based Parsing:
SHIFT -> The
SHIFT -> student
SHIFT -> reads
SHIFT -> the
SHIFT -> book.
Dependency tree created using local transitions.
Graph-Based Parsing:
Possible dependency edges evaluated.
Best global dependency tree selected.
```

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________